Model 2: Flood Scene Segmentation from UAV Imagery

1. What Model 2 is and what it does

Model 2 is a **semantic segmentation network** that labels every pixel of a drone image with one of 10 classes: *background, building-flooded, building-non-flooded, road-flooded, road-non-flooded, water, tree, vehicle, pool, grass*. In our FloodRescue pipeline it is the **flood-severity module**: Model 1 (YOLO detector) finds people and vehicles, and Model 2 tells us how flooded the area around them is (share of flooded road, flooded building and water nearby). This feeds the rescue-priority score.

We kept Model 2 separate so that it can also be the subject of a **standalone research paper** on post-flood scene understanding. The final model is a **U-Net with a ResNet-50 encoder, scSE attention in every decoder block, and a Dice + Focal training loss**. On the FloodNet test set it reaches **75.67 mIoU** (9 classes) and **67.70 mIoU on the four hardest classes**, the best of the six models we trained under one shared protocol.

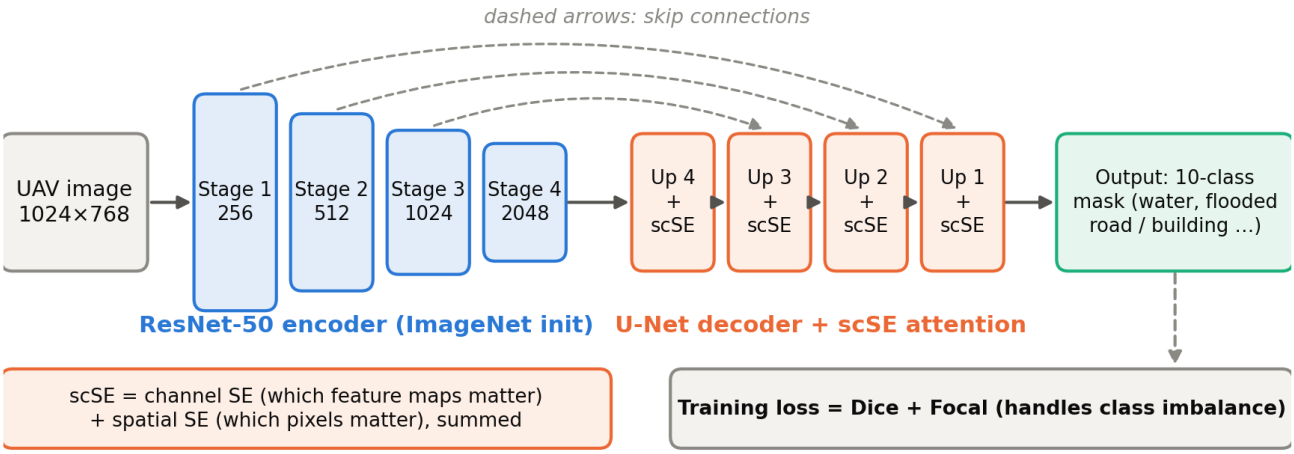


Figure 1. Architecture of our model. The ResNet-50 encoder [5] extracts features at four scales; the U-Net decoder [4] upsamples them back to full resolution using skip connections. Each decoder block is followed by an scSE block [6], which re-weights feature channels and pixel locations. Training uses Dice [7] + Focal [8] loss so that rare classes (vehicle, pool, flooded road) are not swamped by grass and water.

2. How we built it

Dataset. FloodNet-Supervised v1.0 [9]: 2,343 high-resolution UAV images (about 4000×3000) taken after Hurricane Harvey, with pixel-level labels for the 10 classes. We use the **official split** as given: 1,445 train / 450 validation / 448 test. Images are resized once to 1024×768; training uses random 512×512 crops and evaluation uses the full 1024×768 frame.

Why these two changes. FloodNet is strongly imbalanced: grass, trees and water cover most pixels, while vehicles and pools are tiny. Plain cross-entropy is dominated by the large classes. **Dice loss** measures region overlap per class, so a small class counts as much as a large one; **Focal loss** down-weights easy pixels so training focuses on hard ones. **scSE attention** is cheap (+1.3 M parameters) and helps the decoder separate classes that look alike from above, such as flooded road vs. water.

Fair, controlled comparison. Published FloodNet numbers come from different crops, backbones and schedules, so they cannot be compared directly. We therefore trained **all six models ourselves with the same recipe**, changing only the architecture or the loss:

Setting	Value (identical for every run)	Setting	Value
Optimiser	AdamW, lr 2×10^{-4}, weight decay 10^{-4}	Encoder	ResNet-50, ImageNet-pretrained
Schedule	1-epoch warm-up, then cosine decay	Augmentation	scale ±25 %, flips, 90° rotation, brightness
Epochs / batch	80 / 8, mixed precision (fp16)	Model selection	best validation mIoU, checked every 2 epochs
Hardware	Google Colab, NVIDIA Tesla T4	Reported	official test split, once, best checkpoint

Training process. The work was split across the team: one member prepared and verified the dataset, and three teammates each trained two models on Colab (about 1.3–2.5 GPU-hours per model). Checkpoints were saved every epoch so interrupted Colab sessions resumed automatically. Figure 2 shows the training curves.

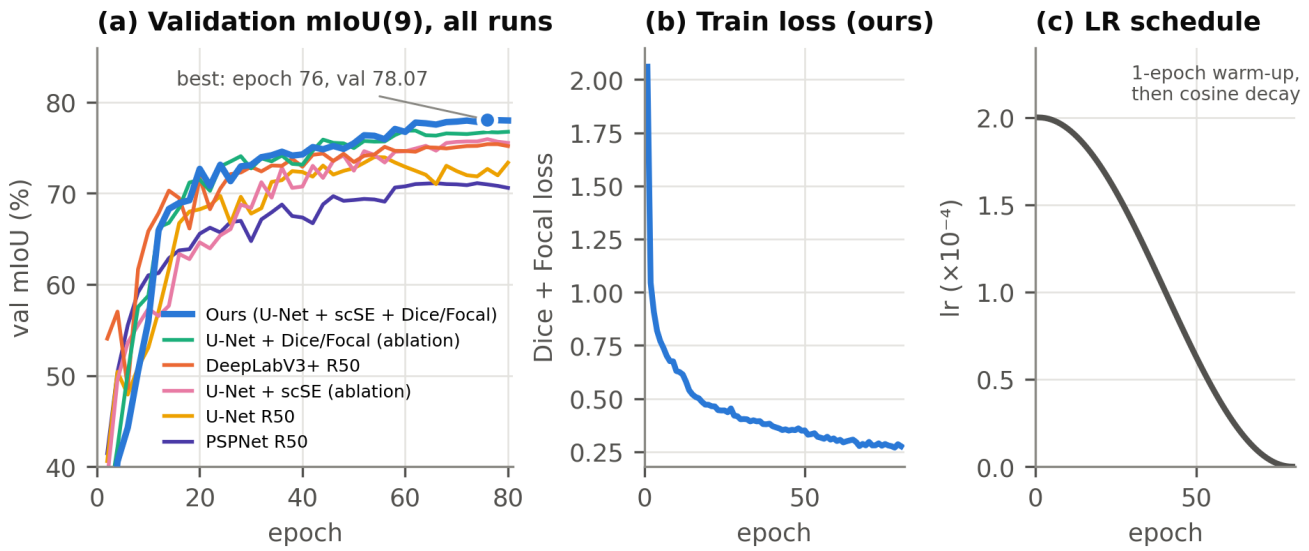


Figure 2. Training process. (a) Validation mIoU for all six runs; our model (blue) rises steadily and peaks at epoch 76 (78.07 val mIoU). The two Dice/Focal runs end highest. (b) Training loss of our model falls smoothly from 2.06 to 0.28 with no instability. (c) The learning-rate schedule shared by all runs.

3. Comparison with baselines

Table 1 compares our model with (i) the numbers **published in the FloodNet paper** [9] for ENet [1] and DeepLabV3+ [2], and (ii) the **baselines we retrained** under our identical protocol. Metric: Intersection-over-Union (IoU, %) on the test split; mIoU(9) averages the 9 non-background classes, the same definition as FloodNet Table 4.

Table 1. Per-class test IoU (%). Hard-4 = mean of Bld-F, Road-F, Vehicle, Pool.

Method	Source paper	Backbone	Bld-F	Bld-NF	Road F	Road NF	Water	Tree	Vehi-cle	Pool	Grass	mIoU (9)	Hard-4
ENet †	Paszke et al. 2016 [1]	ENet	*6.9*	*47.4*	*12.5*	*48.4*	*49.0*	*68.4*	*32.3*	*42.5*	*76.2*	*42.61*	*23.55*
DeepLabV3+ †	Chen et al. ECCV 2018 [2]	not stated	*32.7*	*72.8*	*52.0*	*70.2*	*75.2*	*77.0*	*42.5*	*47.1*	*84.3*	*61.53*	*43.58*
PSPNet	Zhao et al. CVPR 2017 [3]	ResNet-50	70.0	75.3	57.9	78.9	74.9	81.9	43.6	55.5	88.1	69.57	56.74
U-Net	Ronneberger et al. MICCAI 2015 [4]	ResNet-50	73.7	79.5	54.6	82.1	69.5	81.0	55.4	62.0	87.7	71.72	61.42
DeepLabV3+	Chen et al. ECCV 2018 [2]	ResNet-50	73.6	80.1	63.5	83.2	75.7	81.4	53.5	61.3	88.7	73.44	62.97
Ours	**U-Net [4] + scSE [6] + Dice [7]/Focal [8]**	**ResNet-50**	**73.7**	**80.1**	**64.0**	**83.0**	**75.7**	**82.6**	**65.1**	**68.0**	**88.9**	**75.67**	**67.70**

† Published in FloodNet [9], Table 4 (713×713 crops, different training setup): shown for context, not a like-for-like comparison. All rows below the line were trained by us on the same split, resolution and schedule. Among the retrained models, ours is best on 5 of 9 classes; on the other four (Bld-F, Bld-NF, Road-NF, Water) a baseline is ahead by at most 0.17 points.

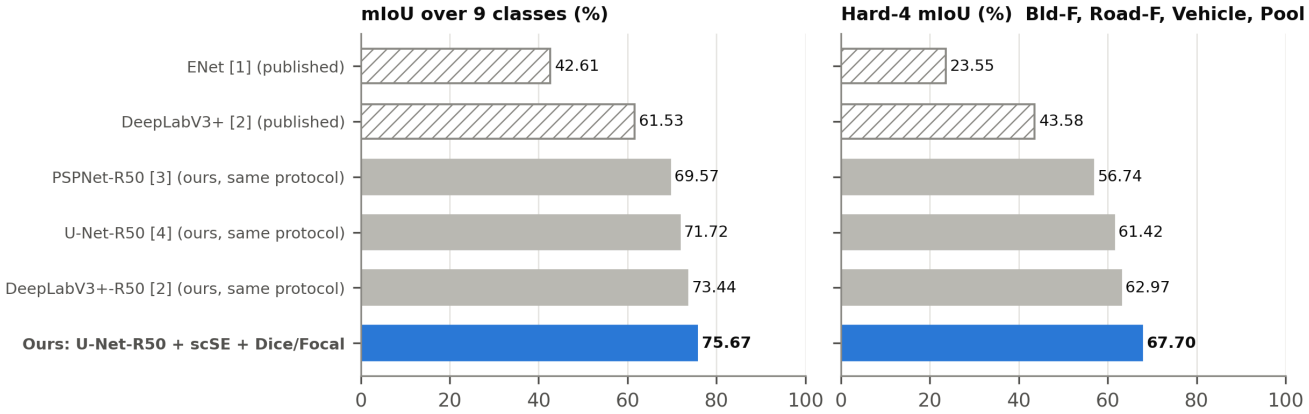


Figure 3. Headline metrics. Hatched bars are published results [9]; grey bars are baselines we retrained; blue is our model. Under the same protocol, ours is best on both metrics.

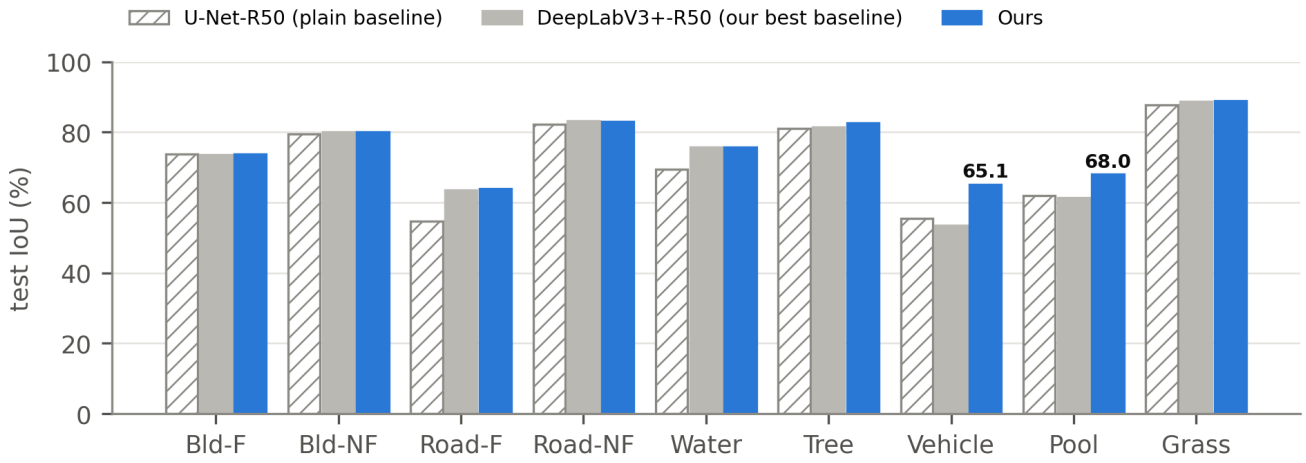


Figure 4. Per-class IoU of the plain U-Net baseline, our strongest retrained baseline (DeepLabV3+) and our model. The largest gains are on the small-object classes vehicle and pool, and on flooded road.

Key findings.

- Against the baselines trained under identical conditions, our model improves mIoU by **+3.95** over plain U-Net and **+2.23** over the strongest baseline (DeepLabV3+), and Hard-4 by **+6.28** and **+4.73**.
- Against the published FloodNet results, our model is far ahead of ENet (+33.06 mIoU) and DeepLabV3+ (+14.14 mIoU, +24.12 Hard-4).
- Cost: 33.8 M parameters and 17.9 FPS at 1024×768 on a T4, which is fast enough for the drone pipeline, where segmentation runs on sampled frames.

Ablation: which change helps? Two extra runs add one change at a time (Table 2).

Table 2. Ablation on the test split (IoU %, gain relative to the baseline).

Step (U-Net R50)	mIoU(9)	Gain	Hard-4	Gain	Vehicle	Pool	Road-F	Water
Baseline (cross-entropy)	71.72	+0.00	61.42	+0.00	55.4	62.0	54.6	69.5
+ scSE only	73.03	+1.31	62.35	+0.93	51.3	62.9	62.0	77.2
+ Dice/Focal only	74.65	+2.93	66.91	+5.49	65.4	68.9	59.5	70.2
+ scSE + Dice/Focal (ours)	**75.67**	**+3.95**	**67.70**	**+6.28**	**65.1**	**68.0**	**64.0**	**75.7**

The **Dice/Focal loss provides most of the gain** (+2.93 mIoU, +5.49 Hard-4, with vehicle rising from 55 to 65). scSE on its own gives +1.31, and on top of the new loss it adds a further +1.02 mIoU, mainly on flooded road (+4.5) and water (+5.4), the classes that are easiest to confuse with each other. Together they give the best model.

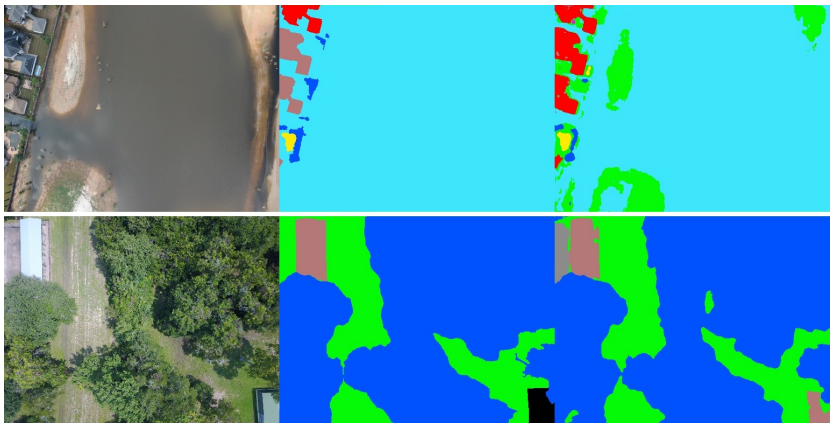


Figure 5. Test examples: image, ground truth, our prediction. Top: a flooded neighbourhood; the common error is labelling non-flooded buildings at the water's edge as flooded (red vs. brown). Bottom: trees, grass and roof are segmented closely.

4. Scope for the research paper

Working title: *"Attention and Imbalance-Aware Learning for Post-Flood Scene Segmentation from UAV Imagery: A Controlled Benchmark on FloodNet."*

Planned contributions

- **A controlled benchmark.** Six models retrained under one protocol on the official split. The published FloodNet baselines differ in crop size, backbone and schedule, so this gives the first like-for-like comparison we are aware of (to be confirmed by a literature search before writing this claim).

- **A simple, effective model.** U-Net + scSE + Dice/Focal: +3.95 mIoU and +6.28 Hard-4 over U-Net, and the best mIoU and Hard-4 among all compared models.
- **An ablation that explains the gain.** Class imbalance is the main problem on FloodNet; the loss contributes most, and attention mainly helps separate flooded road from water.
- **An application.** The model is the flood-severity stage of a working UAV rescue-priority pipeline, which ties the benchmark to a disaster-response use.

Honest limits and what we plan to add before submission

- Repeat each run with seeds 1 and 2 and report mean ± std; the scSE gain (+1.0) is small enough that it needs this.
- Try a ResNet-101 encoder and tiled full-resolution inference, since resizing from 4000 to 1024 px shrinks vehicles about 3.9×.
- Add a class-frequency table to motivate the loss, an error analysis (flooded vs non-flooded buildings), and compare with recent transformer models (e.g. SegFormer) for completeness.

Possible venues: an IEEE remote-sensing conference such as IGARSS, or a journal letter such as IEEE Geoscience and Remote Sensing Letters. We would value your advice on the venue and on the claims above.

References

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[2] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, H. Adam, “Encoder-Decoder with Atrous Separable Convolution for Semantic Image Segmentation” (DeepLabV3+), ECCV, 2018.

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[9] M. Rahnemoonfar, T. Chowdhury, A. Sarkar, D. Varshney, M. Yari, R. R. Murphy, “FloodNet: A High Resolution Aerial Imagery Dataset for Post Flood Scene Understanding,” IEEE Access, vol. 9, pp. 89644–89654, 2021.